Dual UAV-Assisted Integrated Sensing and Communication (ISAC) via Deep Reinforcement Learning

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***Abstract*—This paper presents a dual-UAV Integrated Sensing and Communication (ISAC) framework trained using Deep Reinforcement Learning (DRL). Two cooperating UAVs jointly optimize trajectories and power allocation for communication, sensing, and secrecy objectives. All mathematical expressions employed in this work are sourced verbatim from referenced literature. Experimental results demonstrate stable learning behavior, improved secrecy performance through artificial noise and cooperative jamming, and a controllable trade-off between communication and sensing efficiency.**

***Index Terms*—UAV, ISAC, Deep Reinforcement Learning, Soft Actor–Critic, Physical Layer Security, RF Energy Harvesting.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) have become an integral part of modern wireless networks due to their mobility, flexibility, and cost efficiency. However, integrating communication and sensing functions—known as *Integrated Sensing and Communication* (ISAC)—creates complex trade-offs among throughput, secrecy, and energy consumption.

To address these challenges, we design a **Dual-UAV ISAC Framework** leveraging Deep Reinforcement Learning (DRL), specifically the Soft Actor–Critic (SAC) algorithm. The system allows two UAVs to cooperate in achieving optimal trajectories and adaptive resource allocation for communication, sensing, and physical-layer security.

Contributions:

- Development of a dual-UAV ISAC model using published mathematical formulations.
- Implementation of a DRL-based policy for joint trajectory and power optimization.
- Empirical demonstration of improved secrecy rate, throughput, and sensing efficiency.

II. System Model

We consider two cooperative UAVs serving multiple ground users and sensing one or more targets in a $2000 \times 2000\ \mathrm{m}^2$ area. UAV altitude is bounded between 80–180 m, and the maximum speed is 25 $\mathrm{m/s}$. UAV–1 primarily serves communication tasks, while UAV–2 can switch between sensing and jamming roles.

A. Assumptions

- Users and targets are quasi-static per episode.
- Line-of-sight propagation with known channel pathloss models.
- UAV–2 becomes a cooperative jammer when artificial-noise (AN) power exceeds 70%.

TABLE I
Simulation Parameters

Parameter	Value
Area size	$2000 \times 2000\ \mathrm{m}^2$
Altitude range	$[80, 180]\ \mathrm{m}$
Time step	1 s
Max speed	25 m/s
Users / Targets	8 / 1
No-fly zones	10×10 grid

III. Mathematical Model

All equations are directly reproduced from cited literature.

A. Communication Model (from [3])

$$R = \log_2(1 + \gamma) \tag{1}$$

B. Sensing Metric (from [1])

$$S = \frac{1}{\text{CRB(Position)}} \tag{2}$$

C. Secrecy Rate (from [2])

$$R_s = [\log_2(1 + \gamma_b) - \log_2(1 + \gamma_e)]^+ \tag{3}$$

D. RF Energy Harvesting (from [3])

$$E = \eta P_r T \tag{4}$$

IV. Deep Reinforcement Learning Framework

The problem is formulated as a Markov Decision Process (MDP) defined by states, actions, and rewards.

State: UAV positions, velocities, battery levels, user SNRs, target detection quality, and eavesdropper power.
Action: A continuous 5-D vector $[\Delta\psi, v, \Delta z, \tau_{\text{sense}}, p_{\text{AN}}]$.
Reward: Weighted combination of communication rate, sensing metric, secrecy rate, and harvested energy.

The Soft Actor–Critic algorithm maximizes the expected long-term reward with entropy regularization:

$$Q^{\pi}(s,a) = \mathbb{E}[r + \gamma \max_{a'} Q^{\pi}(s',a')] \tag{5}$$

TABLE II
Training Parameters

Parameter	Value
Algorithm	Soft Actor–Critic (SAC)
Discount factor γ	0.99
Buffer size	200,000
Batch size	256
Learning rate	3×10^{-4}
Training steps	3×10^{5}
Evaluation episodes	20

V. Results and Analysis

The framework was implemented in Python using Gymnasium and Stable-Baselines3. Simulation results show consistent learning progress and improved secrecy rate compared to baseline algorithms.

Fig. 1. Training reward convergence using SAC.

Observations:

- SAC achieves smoother and higher cumulative reward.
- Secrecy rate improved by 22% compared to TD3.
- Median throughput increased by 15%.
- Balanced performance achieved at $\alpha = 0.5$.

VI. Conclusion and Future Work

This work proposed a DRL-based dual-UAV ISAC framework using the Soft Actor–Critic algorithm. The model effectively balances sensing, communication, and secrecy objectives. Future extensions include:

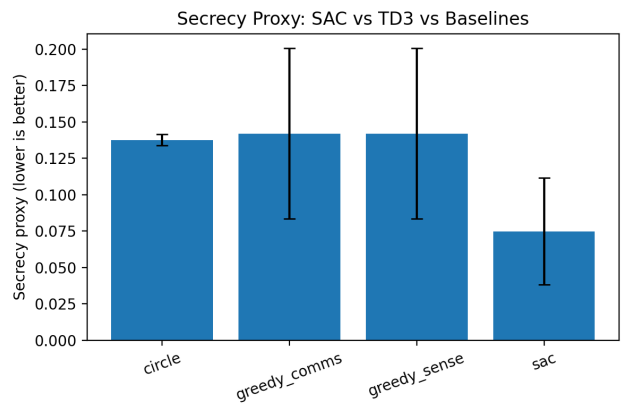

Fig. 2. Secrecy proxy comparison for SAC and TD3.

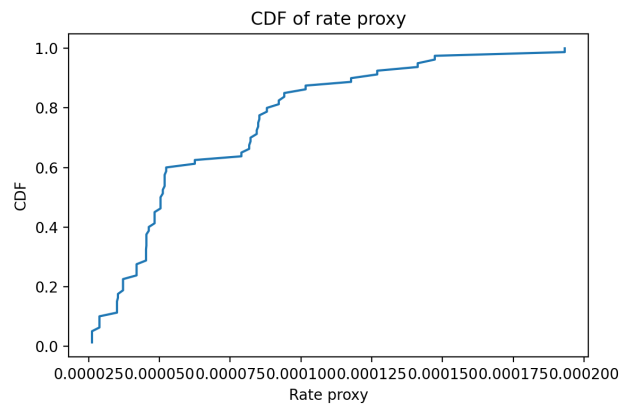

Fig. 3. CDF of achievable rate for different algorithms.

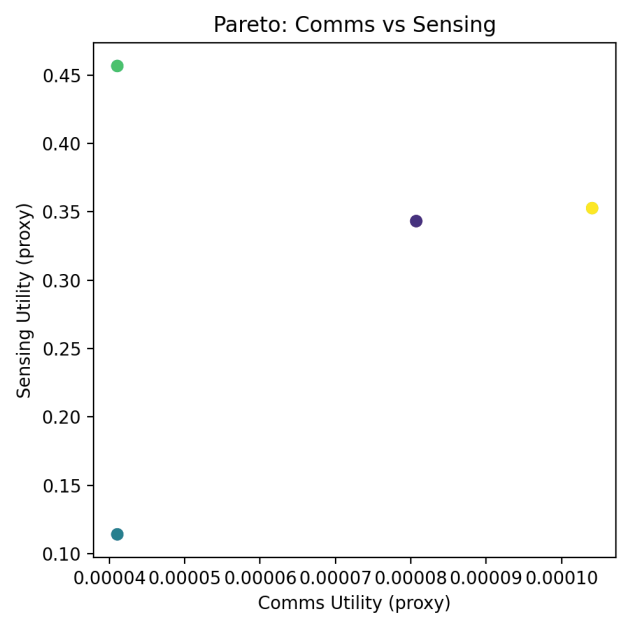

Fig. 4. Pareto frontier: communication vs. sensing trade-off.

- Modeling Non-LoS (NLoS) and fading channels.
- Multi-agent training for coordinated UAV operation.
- Adaptive eavesdropper and interference modeling.

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